



# Part A: Conceptual Understanding (Theory)



## 1. What is Regularization in Machine Learning? Why is it needed?

### Answer:

Regularization is a technique used to reduce **overfitting** by adding a penalty term to the loss function. It prevents the model from becoming too complex and helps it generalize better to new data.

### Why needed?

- Reduces overfitting
- Improves performance on unseen data
- Controls model complexity

## 2. Difference between Ridge Regression (L2) and Lasso Regression (L1).

### Answer:

| Ridge Regression (L2)                  | Lasso Regression (L1)                    |
|----------------------------------------|------------------------------------------|
| Uses squared coefficients as penalty   | Uses absolute coefficients as penalty    |
| Shrinks coefficients toward zero       | Can make coefficients exactly zero       |
| Keeps all features                     | Performs feature selection               |
| Useful when most features are relevant | Useful when some features are irrelevant |

**Summary:** Ridge reduces coefficient values, while Lasso can remove unnecessary features.

### 3. What is Cross-Validation and why is it important?

**Answer:**

Cross-validation is a technique used to evaluate a model by training and testing it on different subsets of data.

**Importance:**

- Provides a reliable estimate of model performance
- Detects overfitting
- Helps in model selection
- Uses data efficiently

### 4. Explain the following cross-validation techniques:

#### a) K-Fold Cross-Validation

**Answer:**

The dataset is divided into **K folds**. The model is trained on K-1 folds and tested on the remaining fold. This process is repeated K times, and the average score is calculated.

**Advantage:** Efficient use of data and reliable evaluation.

#### b) Stratified K-Fold Cross-Validation

**Answer:**

Similar to K-Fold, but each fold maintains the same class distribution as the original dataset.

**Advantage:** Suitable for imbalanced classification datasets.

#### c) Leave-One-Out Cross-Validation (LOOCV)

**Answer:**

A special case of K-Fold where each sample is used as a test set once, and all remaining samples are used for training.

**Advantage:** Maximum use of training data.

**Disadvantage:** Computationally expensive.

#### d) Time Series Split

**Answer:**

Used for time-dependent data. Training is performed on past data and testing on future data while preserving chronological order.

**Use Cases:** Stock prediction, weather forecasting, and sales forecasting.

## **5. Why are tree-based models less sensitive to feature scaling?**

**Answer:**

Tree-based models make decisions using feature thresholds (e.g., Age < 30) rather than distance calculations.

**Therefore:**

- Scaling does not change the order of values.
- Split decisions remain nearly the same.
- Normalization or standardization is usually unnecessary.

**Examples:** Decision Trees, Random Forests, and XGBoost.